

Research Article

Smooth frequency change effects in electrical circuits: explained by the Fresnel integral

Piotr Wach[1]

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Abstract

The paper presents the problem of the relation between ac voltage frequency supplying a circuit and the resulting current frequency in that circuit in the case when the voltage frequency changes smoothly following a $f_v(t)$ function. Contrary to the expectations, the resulting current frequency $f_c(t)$ changes in a different way that this of the voltage. One can say that the current frequency variations are considerably amplified compared with these of the supply voltage. This problem is important because we have to deal with voltage frequency changes that are introduced deliberately, like in case of starting ac motors, and voltage frequency variations that result from external influences, like in case of electrical grids with significant component from wind generating units. Paper solves this problem in analytical way by presenting a relation between supply voltage frequency variations $f_v(t)$ and resulting current frequency $f_c(t)$. It also contains solution of the inverse problem which can find the frequency function of the supply voltage that will result in a required frequency of the current. According to the best author's knowledge, this problem was not reported and examined before and potentially has a number of practical implications, one of which – the induction motor frequency starting – is presented as for example purposes in the paper. The significance of the problem discussed in this paper has still to be examined and established, since frequency variations apply to many engineering branches and until now were counteracted by frequency control without a comprehensive identification of the problem.

Article Highlights

- The supply voltage frequency variations have an important effect on the respective current frequency of the loads. One can forecast that a frequency of the current changes so that it follows exactly the frequency change of the supply voltage. However, in reality the situation looks much different. The frequency of the current changes occur much more intensively than the frequency in the supply voltage, that is the frequency changes of the current amplify the variations of the supply voltage. This phenomenon has neither been examined nor described in technological literature yet, according to best author's knowledge.
- The paper presents, explains and solves the problem in an analytical way. It presents a formula to calculate the frequency of a current when a function of the frequency of the supply voltage is given. The paper also contains a solution of the inverse of the problem described above. There is presented general form of an ordinary differential equation that serves to find a supply voltage frequency function which results in required frequency of the current in a circuit. Presented solutions are general and hold for any lumped parameters circuits: single phase, multi-phase as well as for ac electrical motors supply. In the paper this is demonstrated for single phase circuit supplied by the three exemplary voltage frequency functions as well as for the case of induction motor frequency starting.

✉ Piotr Wach, p.wach@po.edu.pl | [1]Faculty of Electrical Engineering, Automatic Control and Informatics, Politechnika Opolska, Prószkowska 76 Str., 45-758 Opole, Poland.

Keywords Analog circuits · Circuit simulation · Fresnel integral · Frequency modulation · Induction motors starting · Voltage and current frequencies

1 Introduction

In alternating current power applications we often have to do with conditions in which the frequency of a supply voltage is not constant but changes smoothly. This is attributable to a deliberate frequency change, for example during the frequency starting of ac motor drives as well as in the conditions of an external effect acting on a system, and an example of such case is an electric grid in which the influence of wind driven generators is strong. The common expectation is that when the supply voltage frequency variations occur, the current frequency in the respective circuit follows strictly the frequency of the voltage. But it is not the case. The current frequency changes in a different manner than the case of the frequency of the supply voltage, and in a general way one can say the variations in the current frequency amplify the changes of frequency of the supply voltage. For example in a case of voltage frequency increase, the frequency of the current in a circuit increases much more than this of the voltage and the phenomenon of current frequency overshoot is observed. This observation comes the result of the research presented in this paper and it was not, as far as the author knows, presented earlier. It is so probably, because in most applications, when the voltage frequency variations occur, the effort focuses on counteracting them by the voltage frequency control. The problem is related to the Fresnel integral [1–4], which originated in optics, and in the case of electrical power engineering this integral could be applied in some specific voltage frequency functions $f_v(t)$. The outlined problem is solved and presented in this paper in an analytical way. The frequency function $f_v(t)$ of a supply voltage is introduced and the frequency $f_c(t)$ of the resulting current is found. Section 2 shows that for some kind of voltage frequency functions $f_v(t)$ electric current could be solved analytically by use of the Fresnel integral, while generally the current course could be solved by the numerical computation of respective ordinary differential equations. But regardless of current waveform finding, it is possible to find analytical solution for the $f_c(t)$ frequency of the current if only the supply voltage frequency function $f_v(t)$ is a continuous function. This is discussed in Sect. 3. In Sect. 4 the solution of the inverse problem is presented. It is the case when we require some frequency $f_{cr}(t)$ of the current as the output and we need to find what kind of the voltage frequency $f_{vr}(t)$ will provide such an effect. In this section an ordinary differential equation is developed with the purpose of solving this problem. All theoretical considerations are illustrated in the paper by three exemplary functions representing voltage frequency increase: linear, sinusoidal and exponential, but also a case of decrease of the voltage frequency is illustrated. The results are extensively illustrated to demonstrate various rates of frequency increase and to support the results obtained by the analytical solutions. With the purpose of achieving a practical effect, Sect. 5 presents a case study for an induction motor starting by the linear increase of frequency of supply voltage to demonstrate a rotational speed overshoot that happens as the result of 3-phase current frequency overshoot in the stator windings. Finally, Sect. 6 contains discussion of the results in respect to the related problems presented in literature and open fields for further studies, and Sect. 7 presents main conclusions of this research.

2 R,L circuit supplied by changing frequency voltage

A simple *R,L* circuit presented in Fig. 1 forms a sufficient structure to demonstrate all major effects of changing frequency of the voltage supply. The supply voltage is

$$u(t) = U_m \cdot \sin(2\pi f_v(t) \cdot t) \tag{1}$$

Electric current in this circuit, after an assumption is made to adopt zero initial condition for electric current, is calculated by integrating the following differential equation

$$\mathrm{i(t)} = \frac{1}{L}\int_0^t (U_m \sin(2\pi f_v(\tau) \cdot \tau) - Ri(\tau))d\tau \tag{2}$$

To illustrate the problem of the current frequency, the integral of the supply voltage is analyzed, as it plays a key role in solution of (2)

$$\int_0^t U_m \sin(2\pi f_v(\tau) \cdot \tau)\mathrm{d}\tau \tag{3}$$

There is no general solution for integral (3) in standard mathematical functions, but there are solutions for some specific $f_v(\mathrm{t})$, for example for $f_v(\mathrm{t})$ represented by a simple polynomial of t. These solutions employ special functions such as the Fresnel integral. First, three selected exemplary frequency functions will be considered: linear (4), sine-shaped (5) and exponential (6), where $\Delta f = f_k - f_0$ means

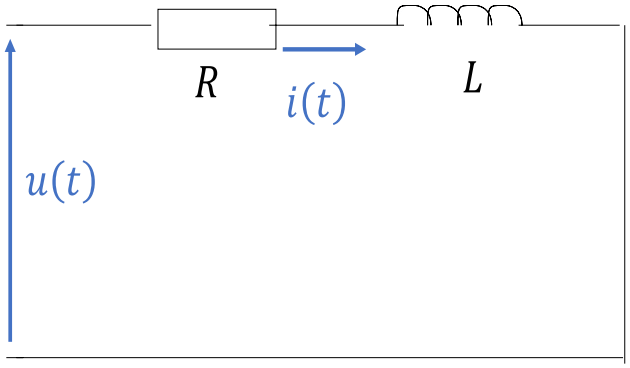


Fig. 1 Electric *R,L* circuit applied for further investigations

change of frequency between initial f_0 and final f_k point of frequency function $f_v(t)$.

The linear frequency function is:

$$\left\{ \begin{array}{cc} f_{v1}(t) = f_0 + \Delta f \frac{t}{T} & for\ t \le T \\ f_k & for\ t > T \end{array} \right\} \tag{4}$$

with: $\Delta f = f_k - f_0$

The sine-shaped frequency function is:

$$\left\{ \begin{array}{cc} f_{v2}(t) = f_0 + \Delta f \sin(\frac{\pi}{2}\frac{t}{T}) & for\ t \le T \\ f_k & for\ t > T \end{array} \right\} \tag{5}$$

And finally the exponential frequency function is:

$$f_{v3}(t) = f_0 + \Delta f\left(1 - e^{-\frac{t}{T}}\right) \tag{6}$$

Plots of these functions are presented in Fig. 2.

For the linear increase function (4) the integral

$$If_{v1}(t) = \int_0^t \left(\sin 2\pi\left(f_0 + \Delta f \frac{\tau}{T}\right)\tau\right) d\tau$$

could be rewritten in a more concise form

$$If_{v1}(t) = \int_0^t sin(a\tau + b\tau^2) d\tau \tag{7}$$

with: $a = 2\pi f_0 \quad b = 2\pi \frac{\Delta f}{T}$

The solution of integral (7) is

$$If_{v1}(t) = \sqrt{\frac{\pi}{2b}}\left[cos\left(\frac{a^2}{4b}\right)S\left(\frac{a+2bt}{\sqrt{2\pi b}}\right) - sin\left(\frac{a^2}{4b}\right)C\left(\frac{a+2bt}{\sqrt{2\pi b}}\right)\right] \quad for\ \ t \le T \tag{8}$$

where: $S(t)$ and $C(t)$ are the Fresnel sine and cosine integrals, respectively [1–4]. From (8) one can compute an

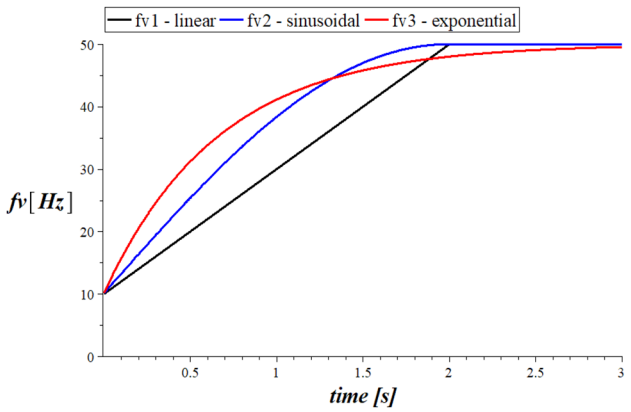


Fig. 2 Voltage frequency functions: linear f_{v1}, sinusoidal f_{v2} and exponential f_{v3}

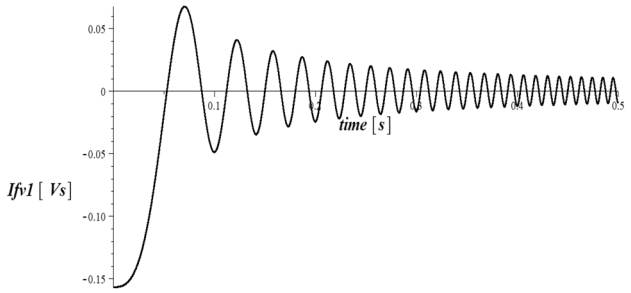


Fig. 3 Solution of the voltage integral (7), in case of linear increase of frequency, from $f_0 = 0\,(\mathrm{Hz})$ to $f_k = 50\,(\mathrm{hz})$, $T = 0.6\,(\mathrm{s})$ by the use of Fresnel integral

integral of voltage for linear increase of frequency, according to f_{v1} function.

Besides, it is necessary to explain here that continuous frequency increase lines in Fig. 4 were derived in a numerical way on the basis of the integral of voltage waveforms, in the manner similar as in Fig. 3, using the formula

$$f = \frac{1}{2(t_{i+1} - t_i)} \tag{9}$$

where $t_{i+1} - t_i$ is the half wave time, that represents the period between consecutive maximum and minimum (or reverse) value of a voltage course. The numerical method for finding an actual current frequency is necessary, because this frequency is entangled in a way that makes its explicit solution impossible. From Fig. 4 one can see that maximum overshoot of the frequency takes place at the end of the line, and it is equal to

$$\Delta f_{dmax} = f_k - f_0 = \Delta f \tag{10}$$

The same result for the linear increase in function $f_{v1}(t)$ will be presented in an analytical form later. The exemplary

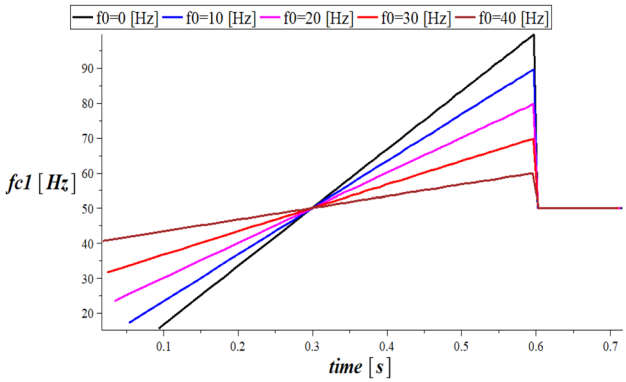


Fig. 4 Frequency increase of voltage integral (8), for $f_0 = 0, 10, 20, 30, 40$ (Hz), $f_k = 50$ (Hz), $T = 0.6$ (s)

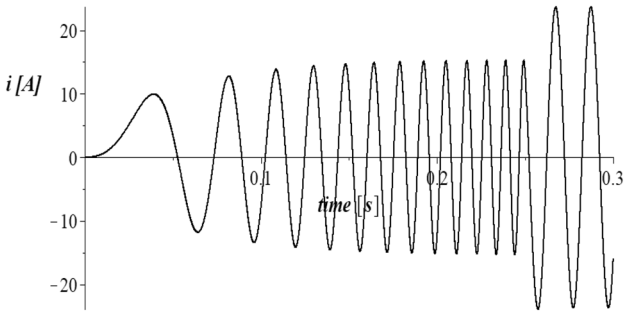


Fig. 5 Electric current in *R,L* circuit for linear voltage frequency increase, according to (4)

shape of the current for linear increase of voltage frequency is presented in Fig. 5.

The effects of the voltage integral and frequency overshoot will be presented in numerical way for the sine-shaped (5) and exponential (6) frequency functions, on the basis of the computed current waveforms in the *R,L* circuit (Fig. 1). This is executed in such a manner since there are no analytical solutions of voltage integral for these kind of functions. In case of the sine-shaped frequency function (5) exemplary result of electric current in the circuit is presented in Fig. 6. Apparently, it is possible to resolve

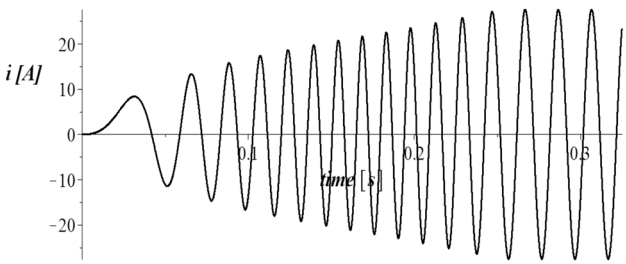


Fig. 6 Electric current in *R,L* circuit for sinusoidal shape of voltage frequency increase

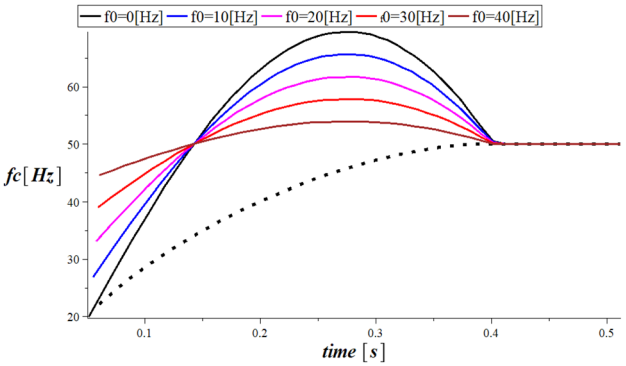


Fig. 7 Frequency of the voltage (the dotted line, $f_0 = 0$ (Hz)) and dynamic frequency of current $f_c(t)$ for sinusoidal increase of voltage frequency from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz)

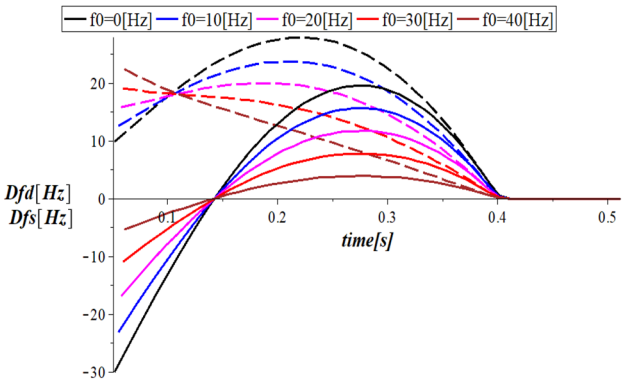


Fig. 8 Dynamic Δf_d – dash lines, and static Δf_s – solid lines, overshoot of frequency of current, for sinusoidal increase of voltage frequency from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz)

these functions (5,6) into power serious, and then apply the Fresnel integral to a limited number of expressions, but this way is not much effective. All computations concerning this circuit are carried out for $R = 10[\Omega]$, $L = 0.03[H]$, $U_m = 230\sqrt{2}[V]$, but it is necessary to explain that the frequency of the computed current is not sensitive to these parameters. It has been tested numerically in a wide range of the time constant $T_p = \frac{L}{R} = 10^{-5} \ldots 10^{5}$ [s]. All numerical computations in this research were carried out by the use of Maple math software, and specifically differential equations were solved by application of the *desolve* procedure intended also for stiff systems.

Figure 7 shows how the frequency of electric current $f_c(t)$ goes over the frequency of voltage (dotted line for $f_0 = 0$) in the *R,L* circuit in case of sinusoidal voltage frequency increase from initial f_0 to the final f_k. The subsequent figure, i.e. Figure 8 shows the dynamic overshoot of current frequency $\Delta f_d = f_c(t) - f_v(t)$, over the $f_{v2}(t)$ voltage frequency, and the static overshoot of current frequency $\Delta f_s = f_c(t) - f_k$, over the final f_k voltage frequency.

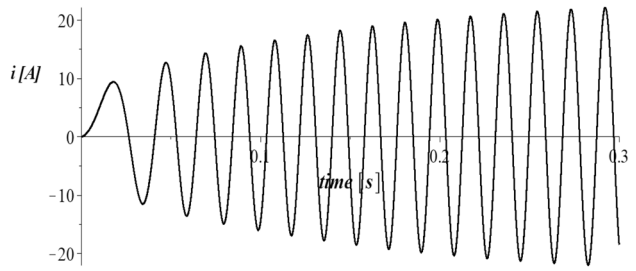


Fig. 9 Electric current in *R,L* circuit for exponential frequency increase of voltage; $T = 0.075$ s

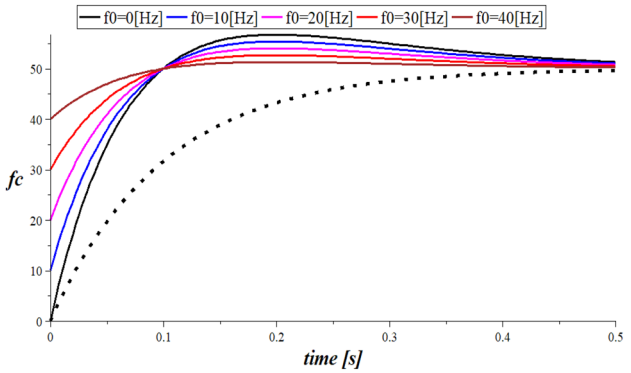


Fig. 10 Frequency of the voltage (dotted line, $f_0 = 0$ (Hz)) and dynamic frequency $f_{c3}(t)$ of current for exponential increase of voltage frequency from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz); $T = 0.075$ s

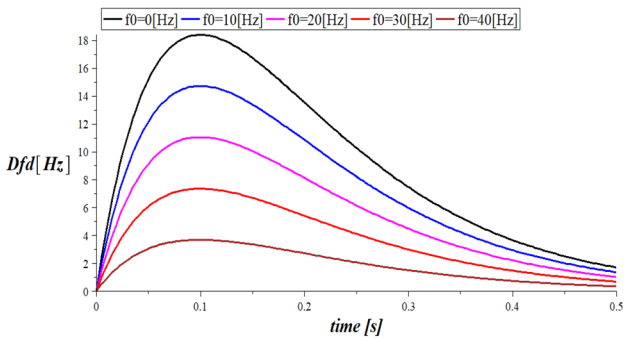


Fig. 11 Dynamic overshoot of frequency of currents Δf_d, for exponential increase of voltage frequency $f_{v3}(t)$, from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz); $T = 0.075$ (s)

The maximum value of overshoot is proportional to difference of $\Delta f = f_k - f_0$, and this proportion will be explained later in the paper. The similar characteristics will be demonstrated for the $f_{v3}(t)$ frequency function, which is exponential. Figure 9 presents electric current in *R,L* circuit, while the supply voltage frequency increases exponentially, and Fig. 10 shows electric current frequency for

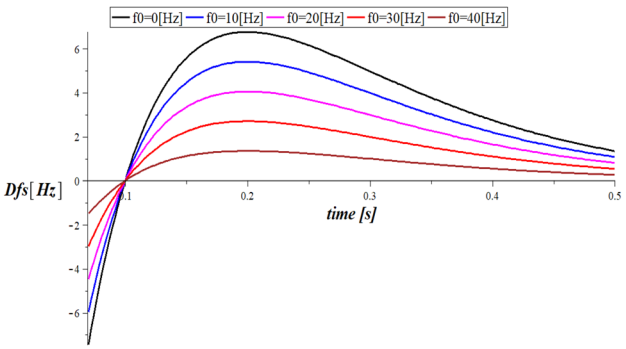


Fig. 12 Static overshoot of frequency of currents Δf_s, for exponential increase of voltage frequency $f_{v3}(t)$ from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz); $T = 0.075$ (s)

$f_0 = 0, 10, 20, 30, 40$ (Hz) with this kind of supply. Figure 11 shows dynamic overshoot Δf_d of current frequency $f_{c3}(t)$, that is surplus of current frequency over voltage frequency $f_{v3}(t)$, and Fig. 12 shows static overshoot Δf_s of current frequency $f_{c3}(t)$, which is the surplus over final f_k frequency.

A summary of the results for these computations are presented in Table 1 for the sinusoidal frequency increase (5) of the voltage, and in Table 2 for the exponential frequency increase (6) of the voltage. The results for the linear voltage frequency increase (4) are not summarized in a table, because in that case the instantaneous frequency of the current is simply doubled in respect to voltage frequency $f_{v1}(t)$ and increases linearly (Fig. 4).

An interesting result from the Table 1 is that the ratio of the maximum dynamic overshoot of current frequency f_{dmax}, in respect to $\Delta f = f_k - f_0$ is constant. Similar observation concerns static overshoot of current frequency $\Delta f_{smax}/\Delta f$. Table 2 contains a collection of quite similar results, but for an exponential increase of the voltage frequency.

The effects of the current frequency changes, in case of decreasing supply voltage frequency, are in principle similar to those when the voltage frequency increase. It means that the frequency of the current $f_c(t)$ goes below the frequency of supply voltage $f_v(t)$ and the overshoot of the current frequency is negative. This is presented in Figs. 13, 14 and Fig. 15 for the initial voltage frequency $f_0 = 60, 50, 40, 30$ (Hz) and the final voltage frequency $f_k = 10$ (Hz). Figure 13 presents the electric current in the circuit for the case of $f_0 = 60$ (Hz), $f_k = 10$ (Hz). It shows very low frequency in the middle of the course and its gradual increase to final $f_k = 10$ (Hz). A collection of graphic results for this example of decreasing voltage frequency along the exponential $f_v(t)$ curves are presented in Fig. 14 and in Fig. 15, to show how much voltage frequencies exceed those of the current frequencies. Numerical results for this case, are presented in Table 3,

Table 1 Characteristics of current frequency in the case of sinusoidal increase of voltage frequency

f_0 (Hz)	$\Delta f = f_k - f_0$ (Hz)	f_{dmax} (Hz)	Δf_{dmax} (Hz)	Δf_{smax} (Hz)	$\Delta f_{dmax}/\Delta f$ (-)	$\Delta f_{smax}/\Delta f$ (-)
0	50.0	69.8	29.7	19.8	0.58	0.390
10	40.0	66.0	22.5	15.8	0.563	0.395
20	30.0	62.0	16.9	11.8	0.563	0.396
30	20.0	56.1	11.3	7.5	0.565	0.375
40	10.0	54.8	5.5	3.8	0.550	0.380

f_{dmax} – maximum of current frequency; Δf_{dmax} – maximum dynamic overshoot of current frequency; Δf_{smax} – maximum static overshoot of current frequency

Table 2 Characteristics of current frequency in the case of exponential increase of voltage frequency

f_0 (Hz)	$\Delta f = f_k - f_0$ (Hz)	f_{dmax} (Hz)	Δf_{dmax} (Hz)	Δf_{smax} (Hz)	$\Delta f_{dmax}/\Delta f$ (–)	$\Delta f_{smax}/\Delta f$ (–)
0	50.0	56.7	18.2	6.7	0.364	0.134
10	40.0	55.4	14.7	5.4	0.368	0.135
20	30.0	54.0	11.0	4.1	0.367	0.137
30	20.0	52.8	7.3	2.8	0.365	0.140
40	10.0	51.4	3.7	1.4	0.370	0.140

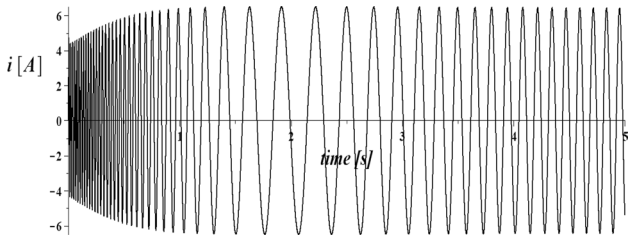


Fig. 13 Electric current in *R,L* circuit for exponential frequency decrease from $f_0 = 60$ (Hz) to $f_k = 10$ (Hz); $T = 0.075$ (s)

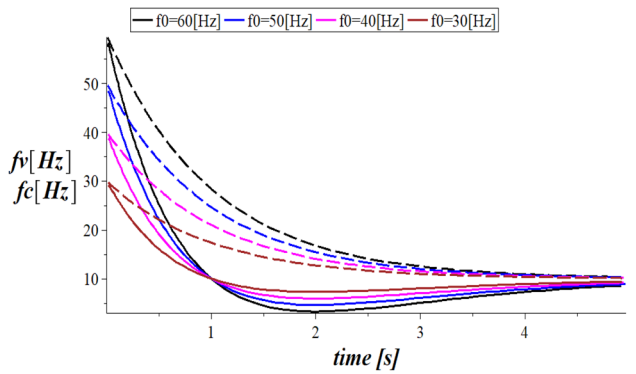


Fig. 14 Presentation of exponential decrease of voltage frequency $f_{v3}(t)$, from $f_0 = 60, 50, 40, 30$ (Hz) to $f_k = 10$ (Hz) – dash lines, and respective frequencies of current $f_{c3}(t)$ – solid lines; $T = 0.075$ (s)

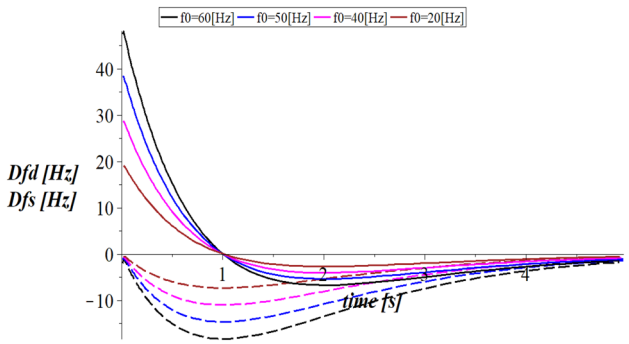


Fig. 15 Overshoot of frequency $f_{c3}(t)$ of current, for exponential decrease ($T = 0.075$ (s)) of voltage frequency$f_{v3}(t)$, from $f_0 = 60, 50, 40, 30$ (Hz) to final $f_k = 10$ (Hz). Static overshoot Δf_s – solid lines, and dynamic Δf_d – dash lines

and we can see that the ratios $\Delta f_{dmax}/\Delta f$ and $\Delta f_{smax}/\Delta f$ have the same values as in case of increasing voltage frequency, presented in Table 2. The interesting result is that for the exponential function of the voltage frequency changes, we receive dynamic overshoot ratio equal to $\Delta f_{dmax}/\Delta f \cong 0.368$, which offers a suitable approximation of the value e^{-1}, and a static overshoot ratio $\Delta f_{smax}/\Delta f \cong 0.135$, which is a good approximation of e^{-2}. These two results confirm the theory presented further in the next paragraph.

The next issue that will be discussed here involves the case in which the final frequency is zero, or negative. One

Table 3 Characteristics of current frequency for exponential decrease of voltage frequency from f_0 to $f_k = 10$ (Hz)

f_0 (Hz)	$\Delta f = f_k - f_0$ (Hz)	f_{dmin} (Hz)	Δf_{dmin} (Hz)	Δf_{smin}	$\Delta f_{dmin}/\Delta f$ (–)	$\Delta f_{smin}/\Delta f$ (–)
60	– 50.0	3.2	– 18.8	– 6.8	0.376	0.136
50	– 40.0	4.8	– 14.9	– 5.2	0.373	0.130
40	– 30.0	6.0	– 11.0	– 4.0	0.367	0.130
30	– 20.0	7.4	– 7.4	– 2.6	0.370	0.130

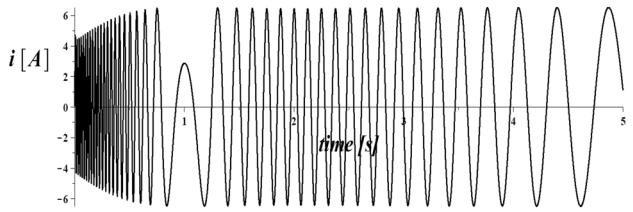


Fig. 16 Electric current in *R,L* circuit for exponential voltage frequency decrease ($T = 0.075$ (s)) from $f_0 = 40$ (Hz) to $f_k = 0$ (Hz)

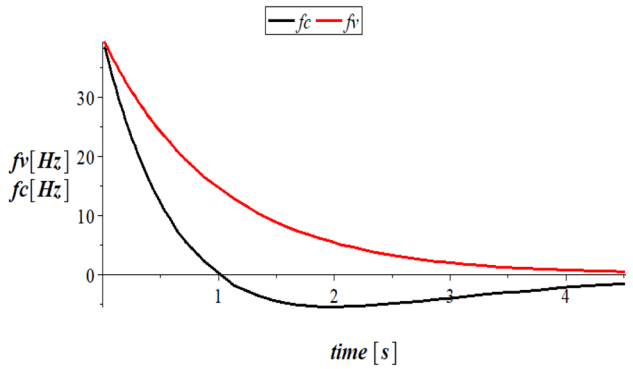


Fig. 17 Frequency of the voltage $f_{v3}(t)$ and dynamic frequency of current $f_{c3}(t)$, for exponential decrease ($T = 0.075$ (s)) of voltage frequency from $f_0 = 40$ (Hz) to $f_k = 0$ (Hz)

example, that deals with such a case is presented in Figs. 16 and 17. There, the voltage frequency $f_v(t)$ decreases exponentially from $f_0 = 40$ (Hz) to $f_k = 0$ (Hz), and current frequency $f_c(t)$, in its dynamic course, goes below zero and then approaches zero from the negative side of the time axis. In a single phase system this does not mean that the frequency is negative, as it just passes zero and changes phase (see Fig. 16). But in the three phase system negative frequency means change of sequence of phases.

3 Analytical solution for dynamic current frequency

The aim of this section is to find and present analytical solution of a current frequency $f_c(t)$ when a supply voltage frequency is given. The differential equation for the *R,L* circuit presented in Fig. 1, in case of linear parameters *R,L* is quite simple

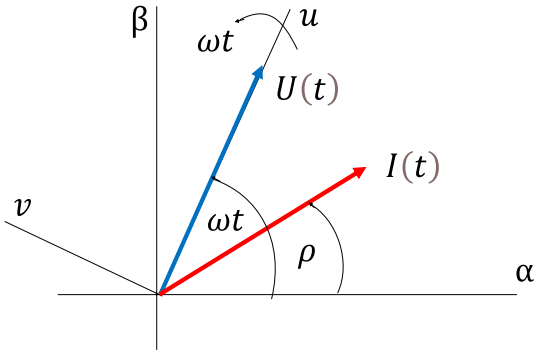


Fig. 18 Illustration of the voltage and current vectors on a complex plane in respect to the stationary frame α,β

$$u(t) = U(t)\sin\left(2\pi f_v(t)\cdot t\right) = Ri(t) + L\frac{d}{dt}i(t) \tag{11}$$

As it was mentioned, analytical solution of (11) does not exist for an arbitrary $f_v(t)$ frequency function, but it is still possible to establish a frequency $f_c(t)$ of the solution (current) in a general form. To find it, let us transform differential Eq. (11) to the complex form, by introducing

$$u(t) = \mathrm{Im}\left\{U(t)e^{j\omega t}\right\} \quad i(t) = \mathrm{Im}\left\{I(t)e^{j\rho}\right\} \tag{12}$$

with: $\omega = \omega(t) = 2\pi f_v(t)\,\rho = \rho(t)$ – current phase angle (see Fig. 18).

and: $U(t)$ – amplitude of the supply voltage $I(t) > 0$ – amplitude of the current.

The differential Eq. (11), after introducing complex variables, takes the form

$$U(t)e^{j(\omega t-\rho)} = RI(t) + L\dot{I}(t) + j\dot{\rho}I(t)L \tag{13}$$

with: $\dot{I} = \frac{d}{dt}I \quad \dot{\rho} = \frac{d}{dt}\rho$

The complex Eq. (13) with the two variables $I(t)$ and $\rho(t)$, can be arranged as a set of two equations

$$\dot{I}(t) = \frac{U(t)}{L}\cos(\omega t-\rho) - \frac{R}{L}I(t) \quad \dot{\rho} = \frac{U}{LI(t)}\sin(\omega t-\rho) \quad I(t) > 0 \tag{14}$$

The solution to this system of differential Eq. (14) allows us to find current phase angle ρ, which is presented in Fig. 19. The small deflections of the ρ curves from a straight line are responsible for the differences between voltage

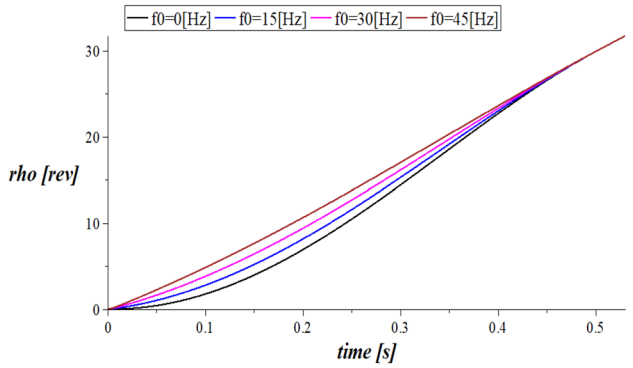


Fig. 19 Rho angle of the rotating electric current vector, obtained as a solution of transformed (14) equations for *R,L* circuit, in case of voltage $f_v(t)$ frequency increase from $f_0 = 0, 15, 30, 45$ (Hz) to $f_k = 60$ (Hz)

and current frequencies. The angle $(\omega t - \rho)$ is quasi-constant, in comparison with the angle ωt, because it changes less than 1 rad during the whole period T of the voltage frequency changes. Even for a short time like T = 0.05 (s), this approximation $(\omega t - \rho) \cong const$ is quite correct, and it is presented in Fig. 20. In fact, this difference denotes the power factor angle φ for the circuit. It changes with frequency, but this change is much limited. On the basis of the equation:

$$\varphi = (\omega t - \rho) \cong \text{const} \tag{15}$$

the time derivative can be calculated, and it gives a very good approximation of the current angular frequency

$$\dot{\rho} = \dot{\omega} t + \omega \tag{16}$$

From (16) we receive

$$f_c(t) = \frac{\dot{\rho}}{2\pi} = f_v(t) + t\frac{d}{\mathrm{d}t} f_v(t) \tag{17}$$

which is the solution for the current frequency.

Applying (17) we can find current frequency for each of the three specific cases of supply voltage frequency $f_g(t)$. For the linear increase (4) it is

$$\left\{ \begin{array}{cc} f_{c1}(t) = f_0 + 2 \cdot \Delta f \frac{t}{T} & for\ t \le T \\ f_k & fort > T \end{array} \right\} \tag{18}$$

– so the dynamic overshoot of current frequency is $\Delta f \frac{t}{T}$, and its maximum is Δf. For the sinusoidal increase (5) of voltage:

$$\left\{ \begin{array}{cc} f_{c2}(t) = f_0 + \Delta f \left[\begin{array}{c} \sin\left(\frac{\pi}{2}\frac{t}{T}\right) + \\ + \frac{\pi}{2}\frac{t}{T} cos\left(\frac{\pi}{2}\frac{t}{T}\right) \end{array} \right] & for\ t \le T \\ f_k & for\ t > T \end{array} \right\} \tag{19}$$

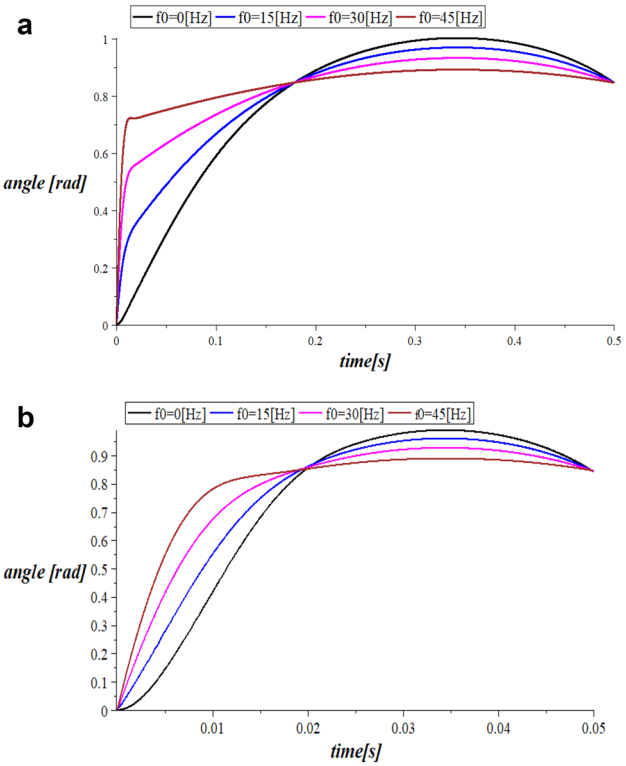


Fig. 20 Results of solution of the equation $\varphi = (\omega t - \rho)$ for the sine-shaped voltage frequency $f_{v2}(t)$ increase from $f_0 = 0, 10, 20, 30, 40$ (Hz) to $f_k = 50$ (Hz). The period of increase is: T = 0.6 (s) or **a**, and T = 0.05 (s) for **b**

- and the maximum of the dynamic overshoot of current frequency is $\Delta f_{dmax} \cong 0.564 \Delta f$ for $t = 2.15375 \frac{T}{\pi}$

- maximum of the static overshoot is: $\Delta f_{smax} \cong 0.392 \Delta f$ for $t = 1.720667 \frac{T}{\pi}$.

Finally, for exponential increase (6) of voltage frequency:

$$f_{c3}(t) = f_0 + \Delta f \left(1 - \frac{T - t}{T} e^{-\frac{t}{T}}\right) \tag{20}$$

– and the maximum of the dynamic overshoot of current frequency is $\Delta f_{dmax} \cong \Delta f e^{-1}$ for $t = T$

- maximum of the static overshoot is $\Delta f_{smax} \cong \Delta f e^{-2}$ for $t = 2T$.

All these analytical results are derived from Eq. (17), and in that sense they are accurate, but we have to remember that Eq. (15) is not accurate, as it assumes that the phase shift φ is constant. Hence, in fact, the Eq. (15) constitutes the assumption for the theory presented in paragraphs 3 and 4. But in practice, the results based on the computer simulation using the differential equation in (11) and results obtained from Eqs. (17) to (19) do not differ noticeably.

4 Inverse problem: finding voltage frequency function f_{vr} for required current frequency f_{cr}

When a required current frequency f_{cr} is given and we look for the f_{vr} voltage frequency that fulfills that task, Eq. (17) becomes an ordinary differential equation that can be applied to solve the inverse problem:

$$\frac{d}{dt}(f_{vr}) \cdot t + f_{vr} = f_{cr} \tag{21}$$

The solution of (21) takes the following form

$$f_{vr} = \frac{1}{t}\int_1^t f_{cr}(\tau)d\tau + \frac{c}{t} \tag{22}$$

where c is an arbitrary constant and the lower limit of the integral can take the form of any nonzero value. Now, let us treat three frequency functions (4)…(6), which originally represented the voltage frequency functions f_v of a linear, sinusoidal and exponential frequency increase, as the required frequency functions f_{cr} for the current in a circuit. By the use of (22) we can find voltage frequency functions f_{vr} to receive the required current frequencies f_{cr}. For the linear increase of current frequency, the result is quite obvious:

$$f_{vr1}(t) = f_0 + \Delta f\frac{t}{2T} \tag{23}$$

as it divides the slope of frequency increase by 2. For the sinusoidal current frequency increase from (22) we receive:

$$f_{vr2}(t) = f_0 - \Delta f\frac{2}{\pi}\frac{T}{t}\cos\left(\frac{\pi}{2}\frac{t}{T}\right) \quad t > 0 \tag{24}$$

and finally for exponential current frequency increase:

$$f_{vr3}(t) = f_0 + \Delta f\frac{T}{t}e^{-\frac{t}{T}} \quad t > 0 \tag{25}$$

For sinusoidal (24) and exponential (25) frequency increase, both f_{cr} and f_{vr} functions are presented in Fig. 21.

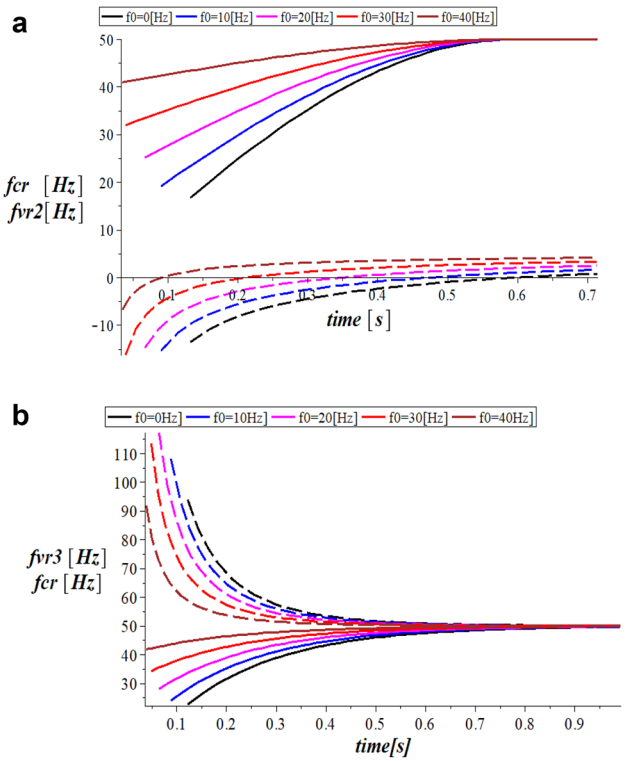


Fig. 21 Illustration of the inverse problem solution: required current frequency functions f_{cr} (as output – solid lines), and voltage frequency functions f_{vr} (as input–dash lines) – for $f_0 = 0, 10, 20, 30, 40$ (Hz) and $f_k = 50$ (Hz). The case presented in Fig. 21. **a** concerns sinusoidal frequency increase ($T = 0.6$(s)), and in Fig. 21. **b** concerns exponential frequency increase ($T = 0.075$(s))

5 Case study for an induction motor starting

To illustrate this problem in more practical conditions, a computer simulation for an induction motor supplied by 3- Level Diode-Clamped Inverter (3-LDCI) is presented in this paragraph. The induction machine: 5.5 [kW], 400 [V], 50 (Hz), 11.5 [A], p = 3 (974 rev/min) gives a typical 3-phase motor design. This drive, with linearly increasing load torque and external inertia, which was four times the rotor's inertia, started with a linear increase of the voltage frequency from 20 (Hz) to final 50 (Hz) in 1.1 [s]. The control of the inverter takes into account the variable f_v frequency throughout the process of computing the actual period of a pulse, with a constant number of pulses, within an accordingly variable 'period' of the supply voltage. Also, the actual voltage space-vector position has been computed accounting for the $f_v(t)$ frequency function in the form that is presented in Fig. 22. The actual value of supply voltage during starting course has been computed using a corrected U/f formula, which takes into account higher output frequency than the one given by the frequency increase curve presented in Fig. 22. The general results of this exemplary simulations are presented in Figs. 23, 24 and 25. In Fig. 24 we can see one of the phase currents with clearly increasing frequency until 1.1 [s] followed by a serious disruption after frequency function switches to the constant frequency of 50 (Hz).

In Fig. 24 we can see that the course of electromagnetic torque runs in a very smooth shape until the sharp disruption occurs after 1.1 [s], when the abrupt frequency change takes place. The maximum negative torque value, about -230 [Nm], is 4.15 multiple of the nominal torque value, which

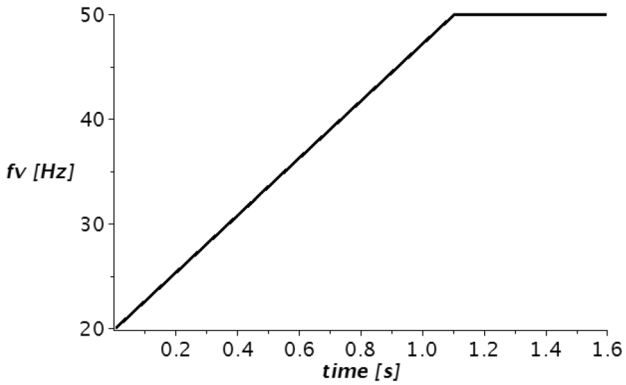


Fig. 22 Linear ramp of voltage frequency increase during induction motor drive starting

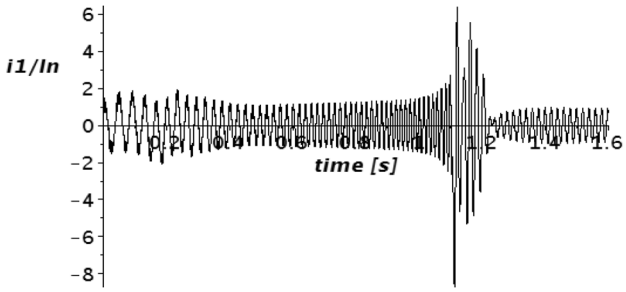


Fig. 23 Phase current of the induction motor during the frequency starting with the linear frequency increase

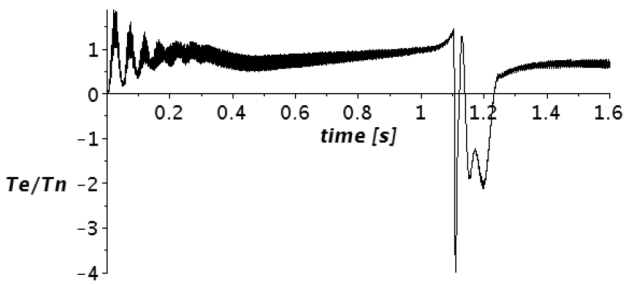


Fig. 24 Electromagnetic torque of the motor during the frequency starting with the linear frequency increase

is realistic. But we have to consider that the mathematical model of the induction motor applied here does not take into account magnetic saturation, and all results concerning high current values are affected by greater error values. Next, in Fig. 25 rotational speed of the drive clearly shows the effect of the frequency overshoot. The maximum value of speed is about 1550 (rev/min), while the synchronous speed at 50 (Hz) is equal to 1000 (rev/min). According to (18), which concerns linear voltage frequency increase, the final f_c frequency in the presented case should be 80 (Hz), which gives 1600 (rev/min) speed of the magnetic field rotation.

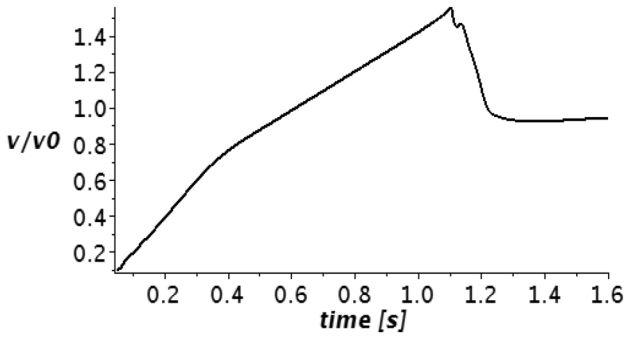


Fig. 25 Rotational speed of the motor during the frequency starting with the linear frequency increase

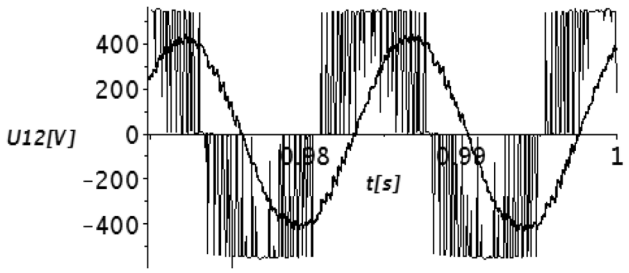


Fig. 26 Motor's clamps voltage and phase current (multiplied by 20) at the final stage of the starting course: f_v =45 (Hz), f_c =72.0 (Hz), n = 1420 (rev/min)

The speed difference between the magnetic field and the rotor speed gives $s = 3.1\%$ of the slip value, which is quite convincing. Finally Fig. 26 presents a short segment of the waveform of the phase current and motor clamps voltage. The motor clamps voltage curves are not drawn very precisely because of the graphic limitations, but the applied basic time step for computations was 2.0E-5 [s], allowing for the high switching frequency. Generally, this example shows that the linear frequency increase is not the best option, due to the considerable current frequency overshoot and a substantial dynamic disruption after the point of the sharp frequency slope change. It seems that the best option should be exponential voltage frequency increase, being a soft curve of change. However, this example does not include any rotor speed regulation during starting course, acting on the voltage frequency curve, which can introduce substantial difference in the operation of the drive.

6 Discussion

The problem discussed here involves a difference between the frequency of the supply voltage and the frequency of the resulting current, and the examples when voltage

frequency changes smoothly, are encountered in many applications and should be examined. Originally the similar problem was found in optics and it was addressed by the Fresnel integral. The Fresnel integral allows to integrate sinusoidal (cosine) functions in cases in which their pulsatance is not constant but it forms a time dependent polynomial. The technological applications of variable frequency in electrical power engineering came much later as a result development of power electronics. Variable frequency in power applications occurs as a result of the deliberate action such as in the case of ac motor drives starting and regulation, but it can also be recorded as a consequence of an external influence acting on a system, like in the case wind power generation. In the first case, when we start ac motor by means of increasing frequency [5–11], an overshoot of rotational speed takes place and it is counteracted by a speed control [12–14]. This problem applies also to zero voltage switching of the invertors that supply variable frequency drives [15–17]. In that case when the frequency of the voltage varies, the switching instant must be controlled by the zero current value, which is different of the zero voltage instant. The second field of applications, in which variable voltage frequency occurs is the case of grids with a large component of the wind power generation [18–20]. An effort is directed at minimizing voltage fluctuations [21–26] but this generally happens without knowledge whether current fluctuations in loads are significantly deeper then these of the supply voltage. The same problem should be investigated from the point of view of standards for power energy generation and consumption [27–29]. Standards should take into account that a deviations of the voltage frequency from nominal value in a grid have more significant impact on load currents than proportional. There are also other fields in which 'frequency coupling' [30–33] is reported as well as current frequency deviations (noise) are counteracted or mitigated [19, 34–36]. These are the areas to study by the respective specialists that should strive to see how much the inconsistency of a supply voltage frequency and resulting current frequency influence an operation of the system. It seems that the phenomenon of voltage and current inconsistency in the case of the supply voltage varying smoothly, was not noticed and discussed until now. All the papers cited above deal with a changing voltage frequency in various applications and its outcomes, but do not notice that the currents frequency changes are not the same as these of the voltage. It seems that all research effort was concentrated on control of voltage frequency [26, 37], (or speed) which overrides the natural tendency of current frequency changes. The studies of the phenomenon reported in this paper should look into the cases of the abovementioned applications, and possibly others, in which the 'overshoot' of current frequency is really meaningful, and find what kind of improvements could be done in particular control systems.

7 Conclusions

The main finding of this paper is that in a circuit supplied by sinusoidal voltage, with the frequency of this voltage varying smoothly to follow some continuous function $f_v(t)$, the frequency $f_c(t)$ of the current changes in a different manner than this of the voltage, and generally this frequency fortifies the tendency of the changes in voltage frequency. It means that if the voltage frequency increases following $f_v(t)$ function, the current frequency $f_c(t)$ increases even higher, and overshoot of current frequency over expected final frequency f_k, is observed. This can be seen from the integral (3) of voltage. This integral could be solved in standard mathematical functions only for some $f_v(t)$ frequency functions, and in such as case Fresnel integral functions are employed. For other $f_v(t)$ functions only numerical solutions exist. The current frequency $f_c(t)$, which is the output frequency of the system, does not depend on the circuit parameters or the period of time T in which these changes occur, as this frequency depends only on the shape of the $f_v(t)$ voltage frequency function. The function $f_v(t)$ has to be a continuous function only, and need not be monotony function, as it was the case for all examples presented here.

The next important conclusion is that the current frequency $f_c(t)$ is the same for a single circuit, multi-phase circuit or induction motor windings, and it depends solely on the integral of $f_v(t)$. This is confirmed in Paragraph 5 for more complex system for the induction motor drive frequency starting. Although there is no general solution of integral (3), it is still possible to establish a formula with the purpose of calculating the frequency of the current (integral of voltage) – see (17). This formula is derived under the assumption that the power factor angle $\varphi = const$, which is not true in general, but under such circumstances, Eq. (15) offers a very good approximation, as it is confirmed even while period T the of frequency change is very short. We can say that if $T > 10/f_k$ then the theory performs very well. The Eq. (17) offers one more possibility – to find such $f_{vr}(t)$, voltage frequency function, that the current frequency follows some required $f_{cr}(t)$ function. It is only necessary to solve the integral in (22).

Declarations

Conflict of interest There was no external funding (outside the University) for this research, no conflict of interests with any individuals,

firms or institutions, and there were no humans or animals involved in the research process.

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